

Data Detective Challenge

1 Dataset Overview

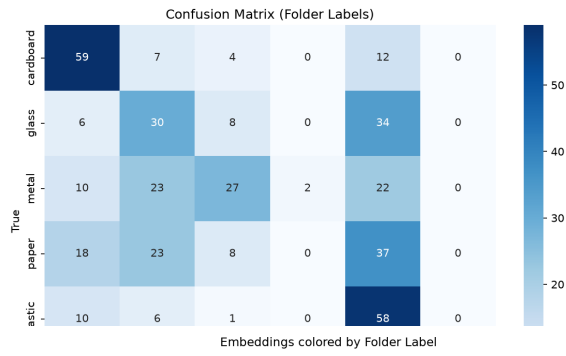
560 images (445 train, 115 test), 6 classes: cardboard, glass, metal, paper, plastic, trash. Filename format: {true_label}_{random_id}.jpg. ResNet-18 from scratch, 10 epochs. Class counts: paper=86, metal=84, cardboard=82, glass=78, plastic=75, trash=40 (2.15x imbalance).

2 Problems Found

Six issues identified:

- **Label Noise (71.5%)**
318/445 images in wrong folder. Filename prefixes also unreliable: model trained on corrected labels achieved 20.22% (random=16.7%), loss stuck at 1.78.
- **Blurry Images (35%)**
157 images with Laplacian variance < 100. Minimum variance: 0.6.
- **Class Imbalance**
Trash (40) vs paper (86) = 2.15x ratio. Per-class accuracy shows trash near zero.
- **Duplicates**
6 exact-MD5 duplicate groups, 11 near-duplicate pHash pairs.
- **Outliers**
22 file-size outliers outside 1.5x IQR. No corrupted images.

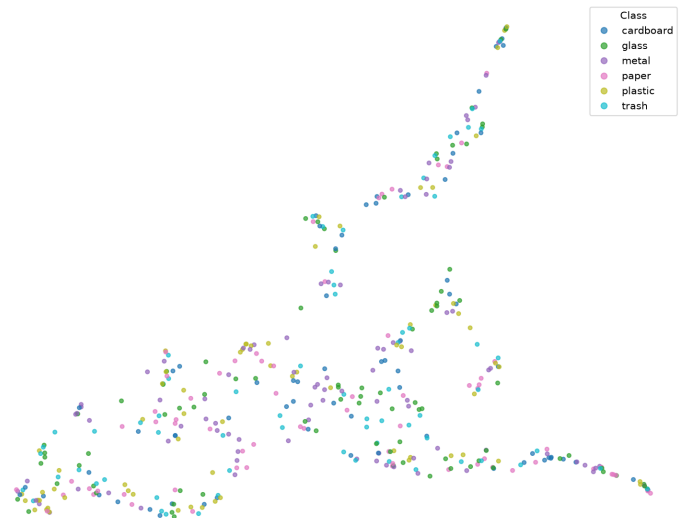
3 Evidence



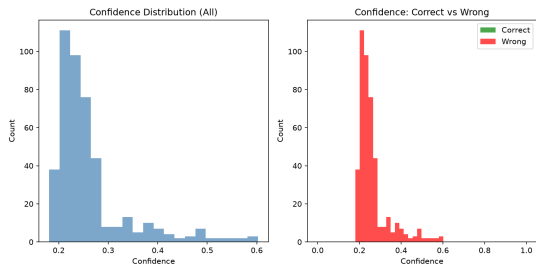
Confusion matrix: predictions concentrate on diagonal for cardboard/glass but scatter for paper -- the class with most folder-label errors.



Embeddings colored by True Label (Filename Prefix)



Left: UMAP by folder label. Right: UMAP by filename prefix.



Confidence distribution: most predictions in 40-60% range. Model lacks high-confidence discrimination, consistent with noisy training labels.

4 Suggested Fixes

- **Re-label dataset [HIGH]**
Both folder labels and filename prefixes unreliable. Use confident learning (CleanLab) or manual verification.
- **Stratified splits [HIGH]**
Stratified sampling to maintain class proportions in splits.
- **Down-weight blurry [MED]**
Lower sample weights for images with Laplacian variance < 100.
- **Augment trash class [MED]**
Aggressive augmentation or class-weighted loss for 2.15x imbalance.
- **Remove duplicates [LOW]**
Deduplicate exact MD5 matches and near-duplicate pairs.
- **Noise-robust training [MED]**
Label smoothing or peer loss for remaining label noise.

5 Expected Impact

Ablation study: same ResNet-18, 10 epochs, two label versions.

Setup	Best Val Acc	Loss
Folder labels (buggy)	44.94%	Decreasing
Filename labels	20.22%	Flat at 1.78
Random chance	16.7%	1.79

Baseline (44.94%) exploits spurious correlations in noisy data. Filename-label model is near random -- neither label source is trustworthy. With proper cleaning, expected 60-70% accuracy (+15-25pp), driven entirely by data quality improvements.